1. Learn SQL in 10 parts

- About 10 minutes each, 100% free
- Databases, grouping, joins, subqueries
- Changing data, window functions, CTEs
- Views and indexes, design, pet shop project
- Start with Part 1 on Beyond School

2. Clean results

- AS gives short names
- DISTINCT removes repeats
- COALESCE replaces NULL with text

```
SELECT name,
COALESCE(email, 'no email yet') AS email
FROM members;
```

name	email
Aisha	aisha@mail.com
Ben	no email yet
Carlos	carlos@mail.com
Dana	no email yet
Eli	eli@mail.com

3. Smart filters

- LIKE: _ = one letter, % = anything
- BETWEEN includes both ends
- IN replaces many ORs
- ORDER BY + LIMIT = top N

```
SELECT title, pages FROM books
ORDER BY pages DESC LIMIT 3;
```

title	pages
Wonder	310
The Hobbit	310
Harry Potter	309

4. Avoid the traps

- Never = NULL, always IS NULL
- Every selected column in GROUP BY (or in COUNT/MAX)
- WHERE filters rows, HAVING filters groups

```
SELECT name FROM members
WHERE email IS NULL;
```

name
Ben
Dana

5. Work safe and fast

- Run SELECT with the same WHERE before UPDATE/DELETE
- EXPLAIN QUERY PLAN shows SCAN vs SEARCH
- CREATE INDEX on columns you search
- WITH (CTE) names your steps